

3. Conceptual Soundness

3.1 Purpose, scope, intended use and restrictions

Before validating mathematics, understand exactly what the model is supposed to do.

For PD:

What probability? Default according to what definition? Over what horizon? For which mortgages? At what point in the mortgage lifecycle?

For LGD:

Loss conditional on what default definition? Over what workout horizon? What recoveries and expenses are included?

Also establish where the model **must not** be used.

This defines the benchmark against which everything else is validated.

3.2 Algorithm/methodology appropriateness

Ask whether the chosen modelling technique is suitable for the problem.

For example, if PD uses logistic regression, determine whether the methodology appropriately represents the binary/default problem and whether assumptions are sufficiently reasonable.

If PD uses gradient boosting or another ML technique, ask whether the additional complexity provides meaningful benefits.

For LGD, consider the characteristics of the target distribution: zeros, tails, bounds, cured cases, incomplete workouts, etc.

The question isn't:

Is XGBoost better than logistic regression?

It's:

Is the selected methodology appropriate for this particular data, target, intended use and risk level?

3.3 Alternative/challenger methodologies

Developers should have reasonable justification for their selected approach.

Compare the final model with:

- incumbent model;
- simpler benchmark;
- alternative statistical model;
- alternative ML model where appropriate.

A materially more complicated model should generally provide a meaningful benefit—performance, stability, risk differentiation, robustness or another justified advantage.

3.4 Model assumptions

Create an assumptions inventory.

Examples could involve:

- default definition;
- independence assumptions;
- recovery assumptions;
- property valuation assumptions;
- treatment of incomplete workouts;
- stationarity assumptions;
- macroeconomic relationships.

For every material assumption:

Assumption → rationale → evidence → sensitivity → consequence if violated

3.5 Limitations and mitigants

Every model has limitations.

Examples:

- limited downturn data;

- small default sample;
- incomplete workouts;
- weak performance in a particular region;
- reliance on proxy property values;
- limited tail observations.

For each limitation determine:

What is the weakness? → How material is it? → What controls reduce the risk? → Is residual risk acceptable?

A mitigant could include an overlay, usage restriction, enhanced monitoring or redevelopment trigger.

3.6 Training methodology

For ML models, review how training was performed.

Check:

- train/validation/test split;
- time-based splitting;
- cross-validation;
- stratification;
- class imbalance;
- preprocessing;
- random seeds;
- reproducibility.

Mortgage data are inherently temporal, so random splitting can sometimes produce misleading results. Out-of-time testing is particularly valuable.

3.7 Parameter estimation and optimization

For statistical models, validate coefficient/parameter estimation.

Check:

- parameter signs;
- magnitude;
- statistical significance where relevant;
- confidence/uncertainty;
- coefficient stability;
- regularization.

For ML models, focus more on optimization procedure, objective/loss function, convergence and stability.

3.8 Hyperparameter tuning and cross-validation

For ML models review:

- which hyperparameters were tuned;
- search ranges;
- tuning method;
- performance metric optimized;
- cross-validation design;
- whether test data remained untouched;
- whether temporal structure was respected.

A major question is whether developers repeatedly used the test set until they obtained good results. That effectively turns the test sample into development information.

3.9 Overfitting and underfitting

Overfitting: Excellent development performance but deterioration on unseen/out-of-time data.

Underfitting: Model is too simple to capture meaningful risk relationships.

Compare:

Training → Validation → Test → Out-of-time

Look at performance differences, stability and segment behaviour.

3.10 Stability and robustness

A sound model shouldn't work only on one lucky sample.

Test performance across:

- time;
- vintage;
- geography;
- lender;
- product;
- LTV band;
- risk band;
- economic environment.

Also test whether small reasonable changes in inputs or samples create disproportionately large changes in results.

3.11 Sensitivity analysis

Sensitivity analysis asks:

If an important assumption/input changes, how does the prediction change?

For mortgage insurance, test important variables such as LTV, credit score, property value, unemployment, interest rates and relevant recovery assumptions.

For LGD, property-value and recovery assumptions can be particularly important.

3.12 Expert judgment

Identify where human judgment entered development.

Examples:

- selecting variables;
- combining categories;
- excluding observations;
- selecting model architecture;
- choosing thresholds;
- adding overlays;
- selecting calibration periods.

For each material judgment determine:

Who made it? Why? What evidence supported it? What would happen under an alternative judgment?

Expert judgment isn't necessarily a weakness. **Undocumented or unsupported expert judgment** is the concern.

3.13 Explainability

Determine whether model behaviour can be understood at a level appropriate to its use.

At minimum understand:

- major risk drivers;
- direction of relationships;
- relative importance;
- reasons for unusual predictions;
- behaviour under changing inputs.

For ML models, techniques such as SHAP may help, but generating a SHAP chart doesn't by itself establish explainability. You should determine whether explanations are **stable, sensible and useful to stakeholders**.

3.14 Economic and business reasonableness

This is broader than statistical performance.

A model can have good AUC/RMSE and still behave strangely.

For example:

- materially higher LTV lowers LGD;
- deteriorating credit quality lowers PD;
- falling property values reduce expected losses.

Those relationships would require strong justification.

Use business intuition, historical experience, segment analysis and sensitivity testing to challenge outputs.

3.15 Third-party components

Identify anything the model depends on that wasn't developed internally:

- external data;
- credit bureau information;
- property valuation services;
- house-price indices;
- vendor models;
- software packages/libraries;
- external APIs.

Understand what happens if the provider changes methodology, data definitions, version or availability.

Third-party components don't eliminate the institution's model risk.

3.16 Reproducibility

This is your final conceptual-development check:

Could an independent qualified validator reproduce the important development results?

Try to reproduce:

Raw data → sample → preprocessing → features → fitted model → predictions → performance results

You don't necessarily need every number to be identical due to legitimate computational differences, but material differences should be explained.

For your validation work, I'd organize all of this into a simple evidence structure:

Requirement → Developer approach → Validation test → Validation evidence → Result → Issue/limitation → Conclusion